High-Temperature Heat Pump Cycle Comparison

Project report template

Status: Starting template; simulations and results are pending. The completed report must not exceed 10 pages.

1 Objective and assumptions

Compare baseline cycle S0 with the assigned advanced cycle (S1–S5). Document the working fluid, boundary conditions, component efficiencies, property method, and modelling assumptions from the assignment.

2 Cycle models

Describe the baseline and selected advanced cycle, including cycle diagrams, state definitions, and mass and energy balances. Implement both models and all analyses in `main.py`.

3 Performance comparison

Report heating coefficient of performance, heat output, compressor power, and any other required metrics. State the common comparison basis and check energy balance closure. No calculated results are available in this template.

4 Parametric analysis

Define the parameter ranges, explain physical constraints, and discuss the resulting performance trends. Save generated plots in `report/figures/` and include selected figures here.

5 Conclusions and references

Summarize the supported findings, limitations, and sources after completing the calculations.